

Case Study 1

1. Mean = 9.11, Median = 7, Mode = 5, 8.
2. Range = 26.
3. Variance = 56.99, Standard Deviation = 7.55.
4. Outlier = 30.
5. Median best represents the typical salary because the outlier increases the mean.

Case Study 2

1. Mean = 150.5 ms, Median = 133.5 ms.
2. Range = 180 ms.
3. Variance = 2558.05, Standard Deviation = 50.58.
4. The data is right-skewed because of the high value 300.
5. Median is more useful because it is less affected by the extreme value.

Case Study 3

1. Q1 = 49.0, Q2 = 53.5, Q3 = 61.0.
2. IQR = 12.0.
3. Upper fence = 79.0; Outlier = 95.
4. Mean = 57.00, Median = 53.5.
5. The outlier increases the mean much more than the median.

Case Study 4

1. Mean = 25.0, Median = 24.5, Mode = 22.
2. Variance = 22.80, Standard Deviation = 4.77.
3. Coefficient of Variation = 19.10%.
4. Standard deviation shows how much daily orders vary from the average.
5. The data is moderately variable.

Case Study 5

1. Mean Error: Model A = 4.00, Model B = 4.62.
2. Variance: Model A = 1.50, Model B = 8.73.
3. Standard Deviation: Model A = 1.22, Model B = 2.96.
4. Model A is more consistent because it has lower variance and standard deviation.
5. No. A lower mean error alone is not enough; consistency and error distribution are also important.